Lecture plan · for the lecturer

A picture is a grid of numbers

University, vision 1

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Read a picture as one or three matrices.
- Compute one filter output by hand.
- Say what edge, blur and sharpen filters do.

Activity: By hand, then the lab

1. Warm-up (5 min): Ask what a filter “sees”, and write the answers on the board.
2. By hand (15 min): Pairs compute one Upright-edges output on a 3×3 patch of 0s and 1s.
3. Lab (15 min): One phone between two: each pair says one nine-term sum aloud, then presses “All at once” and tries Blur and Sharpen.
4. Discuss (5 min): Cross out each board answer that names an object, and write the sum instead.
5. Exit (5 min): One sentence from each student: what does a filter mix?

Discussion

- What is the filter mixing?

Exit ticket

- A filter does not know the name of what?

Try it in your next lecture
Collect the sentence “it knows it is a door” and answer it with the nine multiplications.

Seminar sheet · for students

A picture is a grid of numbers

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Name _____ Group _____ Date _____

1 **A grey picture on a computer is…**

◯ A grid of numbers ◯ A list of object names ◯ One brightness value for the whole photo

2 **A 3×3 filter already knows it is looking at a face.**

◯ True ◯ False

3 **Match the filter.**

Upright edges	• • • •	Keep the strongest in a small block
Blur	• • • •	Walls and door sides
Max pooling	• • • •	A pixel averaged with its neighbours

4 **Order one filter step.**

☐ Write that sum as the new pixel

☐ Lay the 3×3 on the picture

☐ Multiply and add the nine cells

Answers · for the lecturer

A picture is a grid of numbers

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1 **A grid of numbers**

Each cell is a brightness. Colour is three such grids.

2 **False**

It only mixes each pixel with its neighbours.

3 **Upright edges — Walls and door sides**
Blur — A pixel averaged with its neighbours
Max pooling — Keep the strongest in a small block

Convolution mixes neighbours. Pooling throws detail away.

4 **Top to bottom: 3, 1, 2**

The new picture is those sums, one per position.

For you to think about

Where will you stop a student who opens a camera?